SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Short Answers

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Short Answers
Weightage:	30 % of CIA	**Total Marks:**	100 Marks
CO1 Statement:	*Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
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Code of Conduct

I **Subish A G 192312139** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: subish

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1. A telecom operator wins a spectrum auction for 45 MHz. Company policy sets aside 25 kHz for every simplex channel. The planning team wants to know how many full-duplex voice channels this spectrum can support.

Answer: 900 full-duplex channels

Solution: Simplex channels = 45 MHz / 25 kHz = 45,000 / 25 = 1800. A full-duplex channel needs 2 simplex channels, so 1800 / 2 = 900.

Q2. A regional carrier runs its network on 560 total voice channels, shared across a 7-cell frequency reuse cluster. During capacity planning, an engineer needs to know how many channels a single cell will typically be allocated.

Answer: 80 channels per cell

Solution: Channels per cell = 560 / 7 = 80.

Q3. A GSM operator sets carrier spacing at 200 kHz and rolls out 100 carriers, each carrying 8 time slots for voice traffic. Determine the total number of simultaneous voice channels this deployment can offer.

Answer: 800 channels

Solution: Total channels = 100 carriers x 8 slots = 800.

Q4. During a site audit, engineers note the base station radiates 80 W while a nearby handset transmits just 4 W. They want to express this power gap as a decibel-based power control ratio (10·log10 of the ratio).

Answer: ≈ 13.01 dB

Solution: Ratio = 80 / 4 = 20; 10·log10(20) = 13.01 dB.

Q5. A drive test captures a serving signal at -60 dBm alongside co-channel interference at -92 dBm. The RF team needs the Carrier-to-Interference ratio (C/I) in dB to assess link quality.

Answer: 32 dB

Solution: C/I = (-60) - (-92) = 32 dB.

Q6. An operator holds 42 MHz of spectrum, assigning 300 kHz per channel, and reuses frequencies across a 7-cell pattern. How many channels does a single cell end up with?

Answer: 20 channels per cell

Solution: Total channels = 42 MHz / 300 kHz = 140. Per cell = 140 / 7 = 20.

Q7. A transmitter delivers 64 W of power to a receiver. Engineers then move the receiver twice as far away, in an environment with a path-loss exponent of 4. What received power should they now expect?

Answer: 4 W

Solution: $P_r \propto d^{-4}$. Doubling the distance reduces power by $2^4 = 16$. New power = 64 / 16 = 4 W.

Q8. Planners want to double a cell's radius while keeping the same SNR at the new edge, in an environment obeying an inverse-square power law. By what factor should transmit power be scaled up?

Answer: 4 times

Solution: Received power $\propto 1/d^2$. Doubling the radius needs a power increase of $2^2 = 4$.

Q9. A network is built with a cluster size of 4, and every cell in the cluster is provisioned with 15 voice channels. What is the total channel count across the entire cluster?

Answer: 60 channels

Solution: Total = 4 x 15 = 60.

Q10. A drive test records a vehicle traveling at 100 km/h while camped on a 900 MHz carrier. Estimate the Doppler shift the receiver experiences.

Answer: ≈ 83.3 Hz

Solution: v = 100 km/h = 27.78 m/s; λ = c/f = 3x10^8 / 900x10^6 = 0.333 m. fd = v/λ = 27.78 / 0.333 ≈ 83.3 Hz.

Q11. A TDMA frame lasting 5 ms is divided into 8 equal time slots. What is the duration of a single slot?

Answer: 0.625 ms

Solution: Slot duration = 5 ms / 8 = 0.625 ms.

Q12. Spectrum planners have 24 MHz of allocated bandwidth and fix carrier spacing at 200 kHz for a GSM rollout. How many carrier frequencies can be created from this allocation?

Answer: 120 carriers

Solution: Carriers = 24 MHz / 200 kHz = 24,000 / 200 = 120.

Q13. A handset transmits at 8 W, but the signal experiences 33 dB of path loss on its way to the base station. Estimate the power the base station receiver actually sees, in dBm.

Answer: ≈ 6.03 dBm

Solution: 8 W = 10·log10(8000 mW) = 39.03 dBm. Received power = 39.03 - 33 = 6.03 dBm.

Q14. A network engineer uses 1 m as the reference distance while assessing a 1900 MHz deployment, where cell sites are spaced 2.5 km apart. Estimate the free-space path loss over that distance.

Answer: ≈ 106 dB

Solution: λ = 3x10^8 / 1.9x10^9 = 0.158 m. PL(1 m) = 20·log10(4π/λ) ≈ 38.0 dB. PL(2.5 km) = 38.0 + 20·log10(2500/1) = 38.0 + 67.96 ≈ 106 dB.

Q15. A cluster consists of 6 cells, each able to carry up to 250 simultaneous calls at peak. What is the cluster's total call-handling capacity?

Answer: 1500 calls

Solution: Capacity = 6 x 250 = 1500.

Q16. A switching subsystem is rated to process 4000 new call requests every second, and calls last 3 minutes on average. How many calls could be active in the system at any one time?

Answer: 720,000 calls

Solution: Active calls = arrival rate x holding time = 4000 x 180 s = 720,000.

Q17. Engineers assign 150 channels to a base station subsystem, each channel occupying 25 kHz. What total bandwidth does this base station consume?

Answer: 3.75 MHz

Solution: Bandwidth = 150 x 25 kHz = 3750 kHz = 3.75 MHz.

Q18. Spectrum planners insert a 40 kHz guard band between each of 30 adjacent frequency bands to prevent interference. How much total bandwidth ends up wasted because of these guard bands?

Answer: 1.2 MHz

Solution: Wasted bandwidth = 30 x 40 kHz = 1200 kHz = 1.2 MHz. (If guard bands are counted only in the 29 gaps between bands, it is 29 x 40 = 1.16 MHz.)

Q19. A vehicle cruises at 90 km/h through a region of 1.5 km-radius cells. Using the cell diameter as the crossing distance, roughly how many handoffs would occur over an hour of travel?

Answer: ≈ 30 handoffs

Solution: Cell diameter = 2 x 1.5 = 3 km. Distance in 1 hour = 90 km. Handoffs = 90 / 3 = 30.

Q20. A carrier's network hierarchy has 8 regional centers, each overseeing 40 base stations, with each base station serving 250 users. What is the network's total user-handling capacity?

Answer: 80,000 users

Solution: Total = 8 x 40 x 250 = 80,000.

Q21. Coverage planners want to blanket a 400 sq km region using hexagonal cells of 400 m radius (approximate cell area as R^2). Roughly how many cells will the rollout require?

Answer: ≈ 2500 cells

Solution: Cell area ≈ R^2 = $(0.4 \text{ km})^2$ = 0.16 km^2. Cells = 400 / 0.16 = 2500.

Q22. An FDMA system allots 30 kHz to each channel, drawing from a total spectrum pool of 6 MHz. How many channels can be carved out of this allocation?

Answer: 200 channels

Solution: Channels = 6 MHz / 30 kHz = 6000 / 30 = 200.

Q23. A market study notes that 1200 subscribers each place 4 calls per hour, averaging 2.5 minutes per call. What total offered traffic, in Erlangs, does this represent?

Answer: 200 Erlangs

Solution: Traffic = (calls per hour x call duration in hours) = 1200 x 4 x (2.5/60) = 200 Erlangs.

Q24. A 2.5 GHz deployment has 40 MHz of bandwidth, and each connected user consumes 400 kHz. What is the maximum number of users the band can serve at once?

Answer: 100 users

Solution: Users = 40 MHz / 400 kHz = 40,000 / 400 = 100.

Q25. A carrier's base stations each cap out at 600 simultaneous users, and the network comprises 1200 such stations. What total user capacity does the network offer?

Answer: 720,000 users

Solution: Capacity = 600 x 1200 = 720,000.

Q26. A paging scheme sweeps through 4 cells at a time across a 200-cell network. In the worst case, how many paging rounds are needed to locate a subscriber?

Answer: 50 rounds

Solution: Rounds = 200 / 4 = 50.

Q27. A core network is engineered for 100,000 simultaneous connections. During a busy period it receives 40,000 call attempts per minute, each lasting 3 minutes on average. What fraction of the offered traffic ends up blocked?

Answer: ≈ 16.7 % (1/6)

Solution: Offered traffic = 40,000 x 3 = 120,000 Erlangs (simultaneous calls). Capacity = 100,000. Blocked = (120,000 - 100,000) / 120,000 = 0.1667 ≈ 16.7 %.

Q28. A network policy lets 15% of its 600,000 subscribers roam onto partner networks each hour. At any given moment, how many subscribers are likely roaming?

Answer: 90,000 subscribers

Solution: Roamers = 0.15 x 600,000 = 90,000.

Q29. A switching center is capable of processing 12,000 calls per second, but is currently handling a peak load of 9,000 calls per second. What percentage of its capacity is being utilized?

Answer: 75 %

Solution: Utilization = 9000 / 12,000 x 100 = 75 %.

Q30. An operator reuses frequencies with a factor of 4 across a network holding 48 MHz of spectrum, with 200 kHz per channel. How many channels land in each cell?

Answer: 60 channels per cell

Solution: Total channels = 48 MHz / 200 kHz = 240. Per cell = 240 / 4 = 60.

Q31. A 900 MHz GSM deployment uses 200 kHz channel spacing across a 30 MHz allocation. How many channels can this spectrum support?

Answer: 150 channels

Solution: Channels = 30 MHz / 200 kHz = 30,000 / 200 = 150.

Q32. Over the course of an hour, a network logs 2500 handoff attempts, 99% of which succeed. How many calls were successfully handed off?

Answer: 2475 calls

Solution: Successful handoffs = 0.99 x 2500 = 2475.

Q33. A mobile handset radiates 12 W, and the link experiences 45 dB of path loss before reaching the base station. What power level does the base station receiver detect?

Answer: ≈ -4.2 dBm

Solution: 12 W = 10·log10(12,000 mW) = 40.79 dBm. Received power = 40.79 - 45 = -4.21 dBm.

Q34. A 7-cell cluster is deployed where each cell can serve up to 900 simultaneous users. What is the cluster's overall user capacity?

Answer: 6300 users

Solution: Capacity = 7 x 900 = 6300.

Q35. A TDMA carrier's frame spans 4.8 ms and is split into 8 equal slots. How long is each individual slot?

Answer: 0.6 ms

Solution: Slot duration = 4.8 ms / 8 = 0.6 ms.

Q36. A mobile device moves at 150 km/h while connected to a 1.8 GHz carrier. What Doppler shift should the receiver expect?

Answer: 250 Hz

Solution: v = 150 km/h = 41.67 m/s; λ = 3x10^8 / 1.8x10^9 = 0.1667 m. fd = v/λ = 41.67 / 0.1667 = 250 Hz.

Q37. A core switch is dimensioned for 250,000 simultaneous calls. It is fielding calls at a rate of 70,000 per minute, each averaging 4 minutes in duration. What share of the offered load gets blocked?

Answer: ≈ 10.7 %

Solution: Offered load = 70,000 x 4 = 280,000 Erlangs. Capacity = 250,000. Blocked = (280,000 - 250,000) / 280,000 = 0.1071 ≈ 10.7 %.

Q38. A hexagonal layout groups 9 cells into a cluster, and each cell can host 600 users. What is the cluster's combined user capacity?

Answer: 5400 users

Solution: Capacity = 9 x 600 = 5400.

Q39. A switching center processes 60,000 calls per second, with 0.4% of them dropped due to network issues. How many calls are being dropped every second?

Answer: 240 calls per second

Solution: Dropped = 0.004 x 60,000 = 240.

Q40. A base station in an 1800 MHz network transmits at 43 dBm. At the edge of its 2 km-radius cell, the received signal measures -87 dBm. What path loss has the signal undergone?

Answer: 130 dB

Solution: Path loss = Pt - Pr = 43 - (-87) = 130 dB (antenna gains ignored).

Q41. A single base station covers 2.5 km² of territory, and the operator plans to deploy 150 identical stations. What total geographic area will the rollout cover?

Answer: 375 km²

Solution: Total area = 150 x 2.5 = 375 km².

Q42. An operator applies a frequency reuse factor of 6 over a 48 MHz spectrum allocation, with each channel taking up 200 kHz. How many channels does each cell get?

Answer: 40 channels per cell

Solution: Total channels = 48 MHz / 200 kHz = 240. Per cell = 240 / 6 = 40.

Q43. A paging system processes 6000 users per cycle, and the subscriber base totals 150,000 users. How many paging cycles are needed to sweep the entire base?

Answer: 25 cycles

Solution: Cycles = 150,000 / 6000 = 25.

Q44. A phone call runs for 9 minutes, and during that time the network hands the call off roughly every 3 minutes. How many handoffs took place over the call?

Answer: 3 handoffs

Solution: Handoffs = 9 min / 3 min = 3.

Q45. A receiver picks up a wanted signal at -85 dBm against a noise floor of -115 dBm. What signal-to-noise ratio, in dB, does this represent?

Answer: 30 dB

Solution: SNR = (-85) - (-115) = 30 dB.

Q46. A 4G cell is serving 250 users simultaneously, each needing a guaranteed 150 kbps. What is the minimum bandwidth the cell must provide to support them all?

Answer: 37.5 Mbps

Solution: Total = 250 x 150 kbps = 37,500 kbps = 37.5 Mbps.

Q47. A base station handles 1500 subscribers, each averaging 4 calls per hour with a 2.5-minute call length. What total traffic, in Erlangs, does this generate?

Answer: 250 Erlangs

Solution: Traffic = 1500 x 4 x (2.5/60) = 250 Erlangs.

Q48. A network reserves 250 separate control channels, each 12 kHz wide, for signaling. What total bandwidth is tied up in control channels?

Answer: 3 MHz

Solution: Bandwidth = 250 x 12 kHz = 3000 kHz = 3 MHz.

Q49. A network is carrying 1500 simultaneous connections, each streaming at 600 kbps on average. What total throughput must the network support?

Answer: 900 Mbps

Solution: Throughput = 1500 x 600 kbps = 900,000 kbps = 900 Mbps.

Q50. A telecom operator wants each tower to cover a 4 km radius (treating coverage as a circle, area = πR^2) and needs to blanket a 350 km² service region. Roughly how many towers will the rollout require?

Answer: ≈ 7 towers

Solution: Area per tower = π x 4^2 = 50.27 km². Towers = 350 / 50.27 = 6.96, rounded up to 7.